

HW4

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1

$$Y = \begin{pmatrix} y_{11} \\ y_{12} \\ y_{13} \\ y_{21} \\ y_{22} \\ y_{23} \end{pmatrix}$$

X is n by num_treatments. Which ‘treatment’ is each response from?

$$X = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix}$$

Z is n by num_blocks. Which ‘block’ is each response from?

$$Z = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}$$

$$u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

$$e = \begin{pmatrix} e_{11} \\ e_{12} \\ e_{13} \\ e_{21} \\ e_{22} \\ e_{23} \end{pmatrix}$$

2

$$G = \begin{pmatrix} \sigma_b^2 & 0 \\ 0 & \sigma_b^2 \end{pmatrix}$$

$$R = \begin{pmatrix} \sigma^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sigma^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & \sigma^2 \end{pmatrix}$$

$$\begin{aligned} ZGZ' &= \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \sigma_b^2 & 0 \\ 0 & \sigma_b^2 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} \sigma_b^2 & 0 \\ \sigma_b^2 & 0 \\ \sigma_b^2 & 0 \\ 0 & \sigma_b^2 \\ 0 & \sigma_b^2 \\ 0 & \sigma_b^2 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{pmatrix} = \\ &\begin{pmatrix} \sigma_b^2 & \sigma_b^2 & \sigma_b^2 & 0 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2 & \sigma_b^2 & 0 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2 & \sigma_b^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 & \sigma_b^2 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 & \sigma_b^2 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 & \sigma_b^2 \end{pmatrix} \end{aligned}$$

Now add R:

$$V(Y) = ZGZ' + R = \begin{pmatrix} \sigma_b^2 & \sigma_b^2 & \sigma_b^2 & 0 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2 & \sigma_b^2 & 0 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2 & \sigma_b^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 & \sigma_b^2 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 & \sigma_b^2 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 & \sigma_b^2 \end{pmatrix} + \begin{pmatrix} \sigma^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sigma^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & \sigma^2 \end{pmatrix} =$$

$$\begin{pmatrix} \sigma_b^2 + \sigma^2 & \sigma_b^2 & \sigma_b^2 & 0 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2 + \sigma^2 & \sigma_b^2 & 0 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2 & \sigma_b^2 + \sigma^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_b^2 + \sigma^2 & \sigma_b^2 & \sigma_b^2 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 + \sigma^2 & \sigma_b^2 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 & \sigma_b^2 + \sigma^2 \end{pmatrix}$$

3

$$K = \begin{pmatrix} 1/2 & 0 & 0 & 1/2 & 0 & 0 \end{pmatrix}$$

$$KY = y_{11}/2 + y_{21}/2 = \bar{y}_{.1}$$

$$V(KY) = KV(Y)K' = K(ZGZ' + R)K'$$

$$\begin{pmatrix} 1/2 & 0 & 0 & 1/2 & 0 & 0 \end{pmatrix} \begin{pmatrix} \sigma_b^2 + \sigma^2 & \sigma_b^2 & \sigma_b^2 & 0 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2 + \sigma^2 & \sigma_b^2 & 0 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2 & \sigma_b^2 + \sigma^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_b^2 + \sigma^2 & \sigma_b^2 & \sigma_b^2 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 + \sigma^2 & \sigma_b^2 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 & \sigma_b^2 + \sigma^2 \end{pmatrix} \begin{pmatrix} 1/2 \\ 0 \\ 0 \\ 1/2 \\ 0 \\ 0 \end{pmatrix} =$$

$$\begin{pmatrix} (\sigma_b^2 + \sigma^2)/2 & \sigma_b^2/2 & \sigma_b^2/2 & (\sigma_b^2 + \sigma^2)/2 & \sigma_b^2/2 & \sigma_b^2/2 \end{pmatrix} \begin{pmatrix} 1/2 \\ 0 \\ 1/2 \\ 0 \\ 0 \\ 0 \end{pmatrix} = (\sigma_b^2 + \sigma^2)/4 + (\sigma_b^2 + \sigma^2)/4 =$$

$$(\sigma_b^2 + \sigma^2)/2$$

$$K_1 = \begin{pmatrix} 1/2 & -1/2 & 0 & 1/2 & -1/2 & 0 \end{pmatrix}$$

$$V(K_1Y) = K_1V(Y)K_1' = K_1(ZGZ' + R)K_1'$$

$$\begin{pmatrix} 1/2 & -1/2 & 0 & 1/2 & -1/2 & 0 \end{pmatrix} \begin{pmatrix} \sigma_b^2 + \sigma^2 & \sigma_b^2 & \sigma_b^2 & 0 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2 + \sigma^2 & \sigma_b^2 & 0 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2 & \sigma_b^2 + \sigma^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_b^2 + \sigma^2 & \sigma_b^2 & \sigma_b^2 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 + \sigma^2 & \sigma_b^2 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 & \sigma_b^2 + \sigma^2 \end{pmatrix} \begin{pmatrix} 1/2 \\ -1/2 \\ 0 \\ 1/2 \\ -1/2 \\ 0 \end{pmatrix} =$$

$$(\sigma^2/2 \quad -\sigma^2/2 \quad -\sigma^2/2 \quad \sigma^2/2 \quad -\sigma_b^2/2 \quad -\sigma^2/2) \begin{pmatrix} 1/2 \\ -1/2 \\ 0 \\ 1/2 \\ -1/2 \\ 0 \end{pmatrix} = \sigma^2/4 + \sigma^2/4 + \sigma^2/4 + \sigma^2/4 = \sigma^2$$

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$$Z = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

What happens to G?

$$G = \begin{pmatrix} \sigma_b^2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sigma_b^2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma_s^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_s^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma_s^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \sigma_s^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \sigma_s^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_s^2 \end{pmatrix}$$

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Block 1: Treatment 1, Treatment 2, Treatment 3 Block 2: Treatment 1, Treatment 2, Treatment 3 Block 3: Treatment 1, Treatment 2, Treatment 3 Block 4: Treatment 1, Treatment 2, Treatment 3

$$y_{ijk} = tmt_i + block_j + error_{ijk}$$

```
treat_means <- c(25, 35, 30)
sig2_e <- 10
sig2_b <- 8
n_treat <- 3
n_block <- 4
n_samp <- 4
u <- rnorm(4, 0, sqrt(sig2_b))

y <- unlist(lapply(1:n_treat, \(i) lapply(1:n_block, \(j) lapply(1:n_block, \(k) treat_means
```